3 Empirical Methods

(for Documenting Anomalies)

- **Asset-pricing models say that only betas matter** – only betas should explain cross-sectional differences in security returns.
- Yet the literature has documented a number of cross-sectional patters in stock (bond) returns that are not explained by risk – hence they are called 'anomalies'.

- **An anomaly exists when:**
  - **A firm characteristic** predicts which stocks will do better than others—the characteristic predicts cross-sectional differences in asset returns. These are *cross-sectional* anomalies.
  - Examples: size effect (small firms outperform large firms), book-to-market effect (value firms outperform growth firms), momentum (past winners outperform past losers), dispersion effect, credit risk effect, ...
  - **The high-minus-low return spread is statistically significant**—stocks that have low values of the characteristic earn significantly different than stocks that have high value of the characteristic.
  - **The high-minus-low return spread is not due to systematic risk**—the higher returns are abnormal.

3.1 Documenting Anomalies

- Usuall documented via *portfolio sorts* and *cross-sectional regressions*.
- **Portfolio sorts** involve sorting all stocks on a particular characteristic and examining if *the returns of the extreme portfolios* are *significantly different*.
- **The cross-sectional regression approach** involves running Fama-MacBeth type *monthly cross-sectional regressions of returns on the characteristic* and examining if the regression coefficient on the characteristic is *significantly different from zero*.
- In both cases, the goal is to check whether a particular *firm-level characteristic predicts future returns in the cross section*.
- For example, if the characteristic is firm size, we want to know whether firms in the smallest decile portfolio earn significantly different returns than firms in the largest decile portfolio. Alternatively, we would run cross-sectional regressions of future returns on size and examine the regression coefficient.
- We need to ensure that the firm-level characteristic is not *just proxying for risk or for another firm characteristic* already documented to have an impact on the cross-section stock returns.

3.1.1 Portfolio Sorts

- **Each month, sort stocks into decile portfolios** based on a lagged firm-level characteristic, $Z_{i,t-1}$. Occasionally quintiles are used.
  $\longmapsto$ Label the portfolios P1 to P10.
  $\longmapsto$ Note that the portfolio composition will likely change from month to month.
- **Compute portfolio returns** for each portfolio for month t: $r_{1,t}, r_{2,t}, \ldots, r_{10,t}$.
  $\longmapsto$ Portfolio returns could be either

$$\textbf{equally weighted:} \qquad r_{p,t} = \frac{1}{N_p} \sum_{i \in p} r_{i,t} \tag{81}$$

$$\textbf{value-weighted} : \qquad r_{p,t} = \sum_{i \in p} w_{i,t-1} r_{i,t} \tag{82}$$

where N_p is the number of stocks in portfolio p; $\sum_{i \in p} w_{i,t-1} = 1$; $p = 1, \ldots, 10$.

- As a result you have a time-series of T **monthly returns for each portfolio**.
- Test $H_0 : \bar{r}_{10} - \bar{r}_1 = 0$ (use the fact that $\sigma^2(\overline{x}) = \sigma^2(x)/T$)

$$t\text{-}stat_{(\bar{r}_{10}-\bar{r}_1)} = \frac{\bar{r}_{10} - \bar{r}_1}{\sigma(\bar{r}_{10} - \bar{r}_1)} = \frac{\frac{1}{T}\sum_{t=1}^{T}(r_{10,t} - r_{1,t})}{\sqrt{\frac{1}{T^2}\sum_{t=1}^{T}(r_{10,t} - r_{1,t})^2}} \tag{83}$$

3.1.2 Cross-Sectional Regressions

- As we have argued, portfolio-based asset pricing tests may conceal the importance of potentially relevant firm-level characteristics.
- We will use *monthly* FM-type CSR based on *individual* security returns.
- Our **null hypothesis is a K-factor version of the APT**, which implies that expected excess returns are solely determined by factor loadings: $\lambda_k \neq 0, c_m = 0$

$$E_{t-1}(R_{it}) - R_{Ft} = c_{0,t} + \underbrace{\sum_{k=1}^{K} \lambda_{kt}\hat{\beta}_{ik}}_{APT} + \underbrace{\sum_{m=1}^{M} c_{m,t} Z_{i,m,t-1}}_{\textbf{anomalies}} + e_{it} \tag{84}$$

where $Z_{i,m,t-1}$ is a lagged firm-level characteristic for security i.

- We could run the regression above using only raw returns and the characteristics:

$$E_{t-1}(R_{it}) - R_{Ft} = c_{0,t} + \underbrace{\sum_{m=1}^{M} c_{m,t} Z_{i,m,t-1}}_{\textbf{anomalies}} + e_{it} \tag{85}$$

- We would **avoid the EIV problem** due to estimated betas, but the **returns will not be risk-adjusted**—the characteristics could be proxying for risk.

- **To test the significance of characteristic m, we use the sample standard deviation of the monthly OLS estimates of c_{mt}:**

The estimates of c are the *averages across time*:

$$\hat{c}_m = \frac{1}{T}\sum_{t=1}^{T}\hat{c}_{m,t} \tag{86}$$

$$t\text{-}statistic(\hat{c}_m) = \frac{\bar{\hat{c}}_m}{\sigma(\bar{\hat{c}}_m)} = \frac{\frac{1}{T}\sum_{t=1}^{T}\hat{c}_{m,t}}{\sqrt{\frac{1}{T^2}\sum_{t=1}^{T}(\hat{c}_{m,t}-\bar{\hat{c}}_m)^2}} \tag{87}$$

  - Like in the Fama-MacBeth procedure, we use the variation of the statistic *over time* to deduce its variation across samples.

How to adjust for risk?

- In the portfolio sort above, we are using *raw returns*.
- It could be that the characteristic is proxying for risk —the high-minus-low return differential is not abnormal but compensation for risk exposure.
- It could be that the characteristic is proxying for some other firm-characteristic already found to affect returns—this is not a new anomaly.
- There are at least three ways we can **check that the anomaly is robust**:

1. **Risk-adjusted portfolio returns:**
   - **Run a TSR of the high-minus-low return differential, $r_{10,t} - r_{1,t}$, on the risk factors:**

$$r_{10,t} - r_{1,t} = \alpha + \sum_{k=1}^{K} \beta_k F_{kt} + e_t \tag{88}$$

   - Obtain **the alpha**—the intercept in the time-series regression.
   - This intercept is the *abnormal* **return of the anomaly:** $H_0 : \alpha = 0$.
   - If it is significant, that characteristic matters beyond what risk can explain.

2. **Risk-adjusted stock returns:** **Brennan, Chordia, and Subrahmanyam (1998)**
   - **For each stock, run TSR of excess returns on the risk factors.**
     The risk-adjusted returns for each stock are the **intercept and residuals** from this time-series regression:

$$r^*_{it} = r_{it} - rf - \sum_{k=1}^{K} \hat{\beta}_{ik} F_{kt} \equiv a_{it} + e_{it} \tag{89}$$

   - Use these risk-adjusted stock returns to calculate portfolio returns once portfolios are formed as before:

$$r^*_{p,t} = \frac{1}{N_p} \sum_{i \in p} r^*_{i,t} \tag{90}$$

   - **What is the difference between method 1 and method 2?**

3. **Characteristic-adjusted returns:**
   - Use the methodology of Daniel, Grinblatt, et al. (1997).
   - Specifically, **each month (June) triple-sort** stocks into $5 \times 5 \times 5$ portfolios based on size, book-to-market, and past returns.
   - Calculate the average return for each of the 125 'benchmark' portfolios.
   - Then **from each stock return, subtract the average return of the 'benchmark' portfolio to which it belongs**.
   - This is the 'characteristic-adjusted stock return':

$$r_{it}^{ca} = r_{it} - \bar{r}_{k1,k2,k3,t} \tag{91}$$

     where $k1, k2, k3$ are the characteristics bucket to which this stock belongs.
   - Use these characteristic-adjusted stock returns to calculate portfolio returns once portfolios are formed as before:

$$r_{p,t}^{ca} = \frac{1}{N_p} \sum_{i \in p} r_{i,t}^{ca} \tag{92}$$

   - Fama and French (2008) (among others) use size and book-to-market adjusted returns to "disect anomalies"—stock returns are characteristic adjusted relative to 2 not 3 characteristics.

Controlling for Risk and Other Characteristics in CS regressions

- There are at least three ways to ensure the impact of the characteristic is robust:
  1. **Controls in CS regression:** including in the CS regressions characteristics that have already been shown to impact the cross-section of stock returns.
  2. **Characteristic-adjusting returns**: DGTW
  3. **Risk-adjusting returns**: Brennan, Chordia, and Subrahmanyam (1998)

3.2 The Daniel, Grinblatt, et al. (1997) method

- Daniel, Grinblatt, et al. (1997) [DGTW] propose a method for **characteristic-adjusting stock returns**.
- Characteristic-adjusted returns are used when examining the **marginal effect of new anomalies**, i.e. the impact of a firm-characteristic on the returns. The goal is to make sure the 'new' anomaly is not just a proxy for an already documented anomaly.
- Characteristic-adjusted returns are used **both for portfolios sorts** and in **cross-sectional regressions**.
- WRDS contains a sample SAS program for obtaining these DGTW returns for size, book-to-market, and momentum adjusted returns. You can create your own to adjust for fewer, more, or alternative characteristics.
- DGTW use this methodology to assess the market timing and stock picking ability of mutual funds (more on this later).

3.2.1 Obtaining characteristic-adjusted returns as in DGTW

- Each June of year t, stocks are **sorted into portfolios based on a sequential $5 \times 5 \times 5$ sort on size, book-to-market, and past 12-month returns**.
  - The **breakpoints** for this sort are based on **NYSE firms only**, even though NYSE, Amex, and Nasdaq stocks are included in the analysis.
  - **Size** is the market cap (shares×price) as of the end of June of year t.
  - **BM** is the ratio of the book value of equity as of the end of fiscal year $t-1$ to the market value of equity as of December of year $t-1$ (industry adjusted).
  - **Past returns** are the 12-month returns ending in May of year t.

- The **benchmark portfolio returns,** $\bar{r}_{k1,k2,k3,t}$, for each bucket from July of year t to June of year $t+1$, are **value-weighted** across the constituent stocks.
  These benchmark portfolios are buy-and-hold portfolios for 12 months.

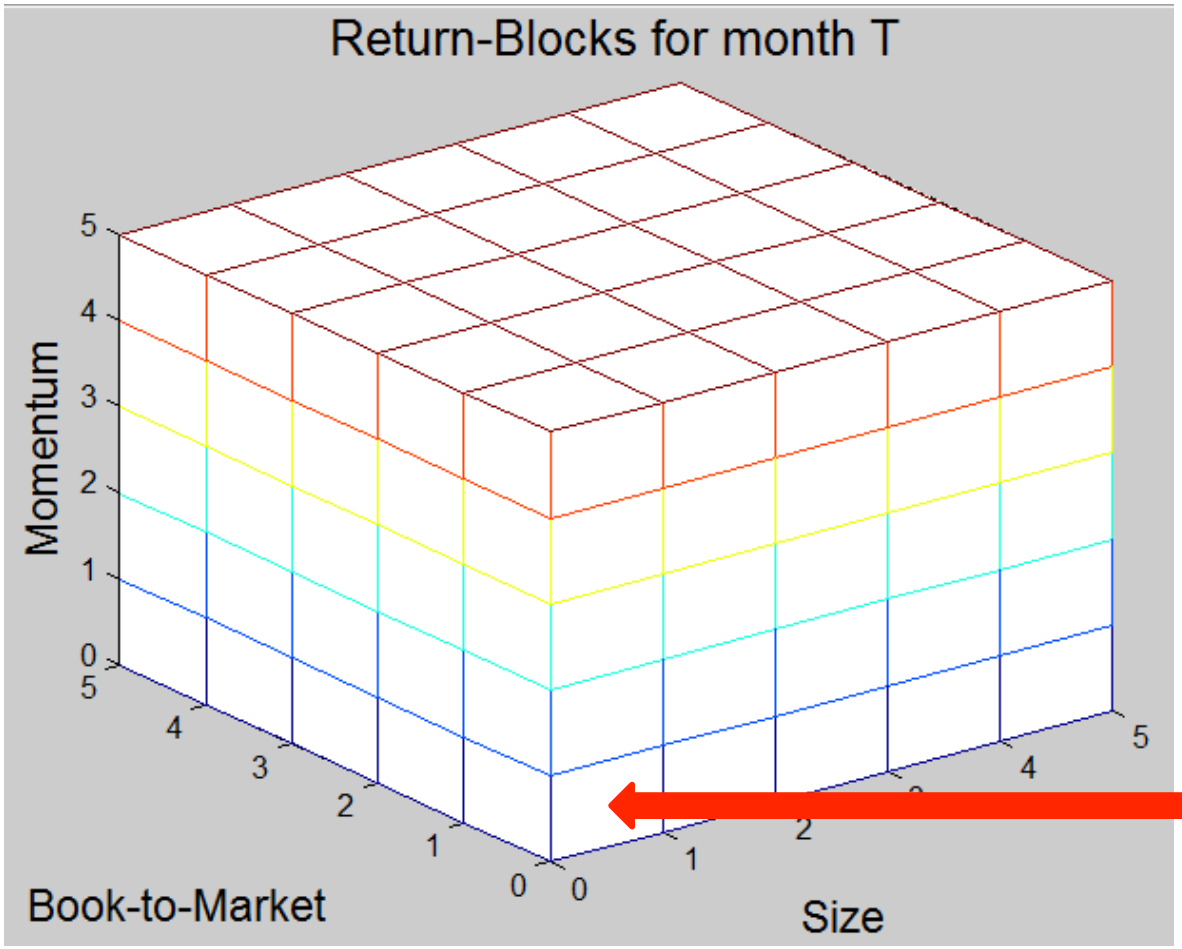
The characteristic-adjusted stock return is:

$$R_{it}^{CA} = R_{it} - \bar{R}_{k1,k2,k3,t} \tag{93}$$

where $k1, k2, k3$ identifies the bucket to which this stock belongs.

- Visualizing the characteristic-adjusted returns (DGTW) method

$$R_{it}^{CA} = R_{it} - \bar{R}_{k1,k2,k3,t}$$



Each return, r(i,t) is compared to its **benchmark** (**matching bucket**)

3.2.2 Evaluating mutual fund performance: by the DGTW method

- Each fund i's return for month t can be decomposed into three components:

$$R_{it} = \sum_{j=1}^{N} w_{j,t-1} R_{j,t} = \underbrace{\sum_{j=1}^{N} w_{j,t-1}(R_{j,t} - R_t^{b_{j,t-1}})}_{CS_t} + \underbrace{\sum_{j=1}^{N}(w_{j,t-1}R_t^{b_{j,t-1}} - w_{j,t-13}R_t^{b_{j,t-13}})}_{CT_t} + \underbrace{\sum_{j=1}^{N} w_{j,t-13}R_t^{b_{j,t-13}}}_{AS_t} \quad (94)$$

- CS measures the **ability to pick stocks within the matching bucket.**
  This is the weighted average 'characteristic-adjusted return' of the fund.
  The benchmark is the matching bucket.
- CT measures the **ability to time the investment style (a bucket).**
  The benchmark are the fund holdings the prior year.
- AS measures the **tendancy to hold stocks with certain characteristics (the bucket).**
  Lagging the weights by a year eliminates returns due to timing the characteristics.

3.3 The Brennan, Chordia, and Subrahmanyam (1998) Method

- Brennan, Chordia, and Subrahmanyam (1998) develop the test to evaluate whether firm-level characteristics impact stock returns.
  1. **First stage: For each asset, run a TSR of excess returns on asset pricing factors.**
  2. **Second stage: Each month, run a CRS of risk-adjusted returns on equity characteristics.**
- Under the null of exact pricing, firm-level characteristics should be statistically insignificant in the cross-section.
- Using risk-adjusted returns, rather than gross or excess returns, is intended to address the finite-sample bias attributable to the EIV problem due to estimating betas in the first-pass TSR.
- Avramov and Chordia (2006) extend the BCS methodology to test pricing models with time-varying factor loadings.

- **Step 1: For each security, estimate betas in a TSR** of excess returns on the observed factors:

$$R_{jt} - R_{Ft} = a_j + \sum_{k=1}^{K} \beta_{jk} F_{kt} + \epsilon_{jt} \tag{95}$$

where and $\hat{\beta}_{jkt}$ is estimated *over the entire sample period.*

  - **The risk-adjusted return** is the intercept and residual from the TSR:

$$R^*_{jt} \equiv R_{jt} - R_{Ft} - \sum_{k=1}^{K} \hat{\beta}_{jk} F_{kt} \equiv \hat{a}_i + \hat{\epsilon}_{jt} \tag{96}$$

  - This risk-adjustment procedure assumes that
    1) the zero-beta return equals the risk-free rate, and
    2) the factor premium = the excess return on the factor (factors is a return).

- **Step 2: Each month, run a CSR of risk-adjusted returns on firm-level characteristics**:

$$R^*_{jt} = c_{0t} + \sum_{m=1}^{M} c_{mt} Z_{mjt-1} + e_{jt}, \tag{97}$$

  where Z_{mjt-1} is the value of characteristic m for security j at time $t-1$, and M is the total number of characteristics.

- Under exact pricing, firm-level characteristics should not explain risk-adjusted return, and $c \equiv [c_m, \ldots, c_M]'$ should thus be insignificant in (97).

- **Each period t, the estimates of c_{mt} are the OLS estimates**.

- **To test the significance of characteristic m, we use the sample standard deviation of the monthly estimates of c_{mt}:**
  - The estimates of c are the *averages across time*:

$$\hat{c} = \frac{1}{T}\sum_{t=1}^{T}\hat{c}_t \tag{98}$$

  - The standard errors are based on the sample variance of the cross-sectional regression estimates and use the fact that $\sigma^2(\overline{x}) = \sigma^2(x)/T$:

$$\sigma^2(\hat{c}) = \frac{1}{T^2}\sum_{t=1}^{T}(\hat{c}_t - \hat{c})^2 \tag{99}$$

3.4 Popular Anomalies

3.4.1 Momentum

- Momentum was well known on Wall Street before the 1990s, but Jegadeesh and Titman (1993) were the first to document it in an academic paper.
- Buying stocks that performed relatively well in the past and selling stocks that performed relatively poorly in the past generate significant positive returns.
- Momentum is profitable for formation and holding periods of 3, 6, 9, 12 months.
- Consider first formation and holding periods of 6 months.
- **The momentum strategy is implemented as follows:**
  1. Each month t, sort stocks into 10 portfolios based on their cumulative returns over months $t-6$ to $t-1$ (*'formation period'*).
  2. Buy the stocks in the top decile (*'winners'*) and sell the stocks in the bottom decile (*'losers'*).
  3. Hold the positions for 6 months, from $t+1$ to $t+6$ (*'holding period'*).

- **The momentum strategy portfolio returns are computed as follows:**

  1. Each month t, and strategy j, compute the strategy-j portfolio returns as *equally* (or *value*-) weighted averages across the constituent stocks.
  2. Since each month, you're holding 6 strategies, compute the *equally* weighted month-t return of the strategy implemented last month and the strategies implemented up to six months earlier.
  3. The momentum profit for month t is the return differential between the month-t return of portfolio P10 (winners) and the month-t return of portfolio P1 (losers).
  4. The result is a time-series of monthly momentum profits, which allows for standard statistical inference based on non-overlapping monthly returns.

3.4.2 Momentum robustness

- Stock return momentum is robust
  - Fama and French (1996) show that momentum is the only CAPM-related anomaly unexplained by the Fama and French (1993) model.
  - In fact, regressing gross momentum payoffs on the Fama-French factors tend to strengthen, rather than explain, momentum profitability.
  - Schwert (2003) demonstrates that the size and value effects in the cross section of returns, as well as the ability of the aggregate dividend yield to forecast the equity premium, disappear, reverse, or attenuate following their discovery.
  - Jegadeesh and Titman (2001, 2002) document the profitability of momentum strategies in the out of sample period.

  - Haugen and Baker (1996) and K. G. Rouwenhorst (1998) document momentum in international markets.
  - Korajczyk and Sadka (2004) find that momentum survives trading costs, whereas Avramov, Chordia, and Goyal (2006) show that the profitability of the other past-return anomaly, namely reversal, disappears in the presence of trading costs.
  - Fama and French (2008) show that momentum is among the few robust anomalies.

- The prominence of the momentum profitability has generated both behavioral and rational theories.
  - Behaviroal: Barberis, Shleifer, and R. Vishny (1998), Daniel, Hirshleifer, and Subrahmanyam (1998), and Hong, Lim, and Stein (2000).
  - Rational: Berk, Green, and Naik (1999) and Johnson (2002).

3.4.3 Momentum interactions

Momentum interactions have been documented at the stock, industry, and aggregate level.

- **Stock level interactions:**
  - Hong, Lim, and Stein (2000) show that momentum profitability concentrates in **small stocks**.
  - Lee and Swaminathan (2000) show that momentum payoffs increase with **trading volume**.
  - X. F. Zhang (2006) finds that momentum concentrates in **high information uncertainty** stocks (stocks with high return volatility, cash flow volatility, or analysts' forecast dispersion) and provides a behavioral interpretation.
  - Avramov, Chordia, Jostova, et al. (2007) document that momentum concentrates in **low rated stocks**. Moreover, the credit risk effect seems to *dominate the information uncertainty and size effects.*

- **Industry level interactions:**
  - Moskowitz and Grinblatt (1999) show that industry momentum subsumes stock level momentum.
  - Grundy and Martin (2001) find no industry effects in momentum.
- **Interactions at the aggregate level:**
  - Chordia and Shivakumar (2002) show that momentum is captured by business cycle variables.
  - Avramov and Chordia (2006) demonstrate that momentum is captured by the component in model mispricing that varies with business conditions.
  - Cooper, Gutierrez, and Hameed (2004) finds that the mean momentum payoff following positive (negative) market return is 0.93% (-0.37%).

3.4.4 Momentum in Other Asset Classes

- Momentum has been documented in **U.S. and international equities, currencies, commodities, sovereign bonds, and residential real estate**. See Jegadeesh and Titman (1993), K. G. Rouwenhorst (1998), Chan, Hameed, and Tong (2000), Okunev and White (2003), Miffre and Rallis (2007), Gorton, Hayashi, and G. K. Rouwenhorst (2008), Asness, Moskowitz, and Pedersen (2013), and Beracha and Skiba (2011).
- Recently, Jostova, Nikolova, et al. (2013) [JNPS] also document it for **U.S. corporate bonds**.
  —While Avramov, Chordia, Jostova, et al. (2013) find that most asset pricing anomalies in equities drive their profitability mostly from the short side of the momentum strategy, corporate bond momentum seems to be driven by the **long side** of the strategy.
  —Moreover, JNPS show that bond momentum is not just a manifestation of equity momentum, but has a strong and robust **bond-driven** component.
  —As with equity momentum, bond momentum is only present among **non-investment grade** firms.

3.5 Other Asset Pricing Anomalies

- See Section 1.2.1 for a review of popular anomalies.
- Examples include **earnings momentum, size, value, dispersion, credit risk, investments, asset growth, and accruals**, among others.
- Fama and French (2008), Lewellen (2015), Avramov, Chordia, Jostova, et al. (2013) examine many of these anomalies in a unified framework.
- Fama and French (2008) find that anomalies are stronger among 'micro' stocks (i.e. those below the 20th NYSE size breakpoints).

- Avramov, Chordia, Jostova, et al. (2013) find that all but the accruals anomaly are present only among **non-investment grade stocks** and are driven only by periods **around downgrades**. All but the value anomaly are driven by the **short side** of the strategy, i.e. from firms that get into financial distress and continue to deteriorate. In contrast, the value anomaly derives its profitability from high credit risk stocks that survive financial distress.
- These results suggest that while these anomalies were documented based on all firms on NYSE, Amex, and Nasdaq, they are only present in a **specific segment** of the market and manifest themselves in **particular periods**.

3.6 Corporate Based Anomalies

- So far we have been dealing with asset pricing anomalies.
- The corporate finance literature has documented a host of other interesting anomalies.
- Some anomalies are suspected to be under-reaction to news some over reaction.
- Here is a list of prominent anomalies
  - Stock Split
  - IPO
  - SEO
  - Dividend initiation and omission
  - Stock repurchase
  - Spinoff
  - Merger arbitrage
- Fama (1998) is a comprehensive source of corporate anomalies.

Discussion

- What could explain why the P10-P1 return differential exists?
- What could explain why it persists?

References for Anomalies

1 Eugene F. Fama and Kenneth R. French. "Common Risk Factors in the Returns on Stocks and Bonds". In: *Journal of Financial Economics* 33.1 (Feb. 1993), pp. 3–56.

2 Narasimhan Jegadeesh and Sheridan Titman. "Returns to buying winners and selling losers: implications for stock market efficiency". In: *Journal of Finance* 48.1 (Mar. 1993), pp. 65–91.

3 Eugene F. Fama and Kenneth R. French. "Multifactor Explanations of Asset Pricing Anomalies". In: *Journal of Finance* 51.1 (Mar. 1996), pp. 55–84.

4 Robert A. Haugen and Nardin L. Baker. "Commonality in the determinants of expected stock returns". In: *Journal of Financial Economics* 41.3 (July 1996), pp. 401–439.

5 Kent Daniel, Mark Grinblatt, Sheridan Titman, and Russ Wermers. "Measuring Mutual Fund Performance with Characteristic-Based Benchmarks". In: *Journal of Finance* 52.3 (July 1997), pp. 1035–1058.

6 Nicholas Barberis, Andrei Shleifer, and Robert Vishny. "A model of investor sentiment". In: *Journal of Financial Economics* 49.3 (Sept. 1998), pp. 307–343.

7 Michael J. Brennan, Tarun Chordia, and Avanidhar Subrahmanyam. "Alternative factor specifications, security characteristics, and the cross-section of expected stock returns". In: *Journal of Financial Economics* 49.3 (Sept. 1998), pp. 345–373.

8 Kent Daniel, David A. Hirshleifer, and Avanidhar Subrahmanyam. "Investor Psychology and Security Market Under- and Overreactions". In: *Journal of Finance* 53.6 (Dec. 1998), pp. 1839–1885.

9 Eugene F. Fama. "Market efficiency, long-term returns, and behavioral finance". In: *Journal of Financial Economics* 49.3 (Sept. 1998), pp. 283–306.

10 K. Geert Rouwenhorst. "International Momentum Strategies". In: *Journal of Finance* 53.1 (1998), pp. 267–284.

11 Jonathan B. Berk, Richard C. Green, and Vasant Naik. "Optimal Investment, Growth Options, and Security Returns". In: *Journal of Finance* 54.5 (1999), pp. 1553–1607.

12 Tobias J. Moskowitz and Mark Grinblatt. "Do Industries Explain Momentum?" In: *Journal of Finance* 54.4 (1999), pp. 1249–1289.

13 Kalok Chan, Allaudeen Hameed, and Wilson Tong. "Profitability of Momentum Strategies in the International Equity Markets". In: *Journal of Financial and Quantitative Analysis* 35.2 (June 2000), pp. 153–172.

14 Harrison Hong, Terence Lim, and Jeremy C. Stein. "Bad News Travels Slowly: Size, Analyst Coverage, and the Profitability of Momentum Strategies". In: *Journal of Finance* 55.1 (2000), pp. 265–295.

15 Charles M.C. Lee and Bhaskaran Swaminathan. "Price Momentum and Trading Volume". In: *Journal of Finance* 55.5 (2000), pp. 2017–2069.

16 Tarun Chordia and Lakshmanan Shivakumar. "Momentum, Business Cycle, and Time-varying Expected Returns". In: *Journal of Finance* 57.2 (2002), pp. 985–1019.

17 John Okunev and Derek White. "Do Momentum-Based Strategies Still Work in Foreign Currency Markets?" In: *Journal of Financial and Quantitative Analysis* 38.2 (June 2003), pp. 425–447.

18 G. William Schwert. "Anomalies and Market Efficiency". In: *Handbook of the Economics of Finance*. Ed. by George Constantinides, Milton Harris, and René Stulz. Simon School Working Paper No. FR 00-21; NBER Working Paper No. W7935. Amsterdam: North-Holland, 2003, pp. 937–972.

19 Robert A. Korajczyk and Ronnie Sadka. "Are Momentum Profits Robust to Trading Costs?" In: *Journal of Finance* 59.3 (2004), pp. 1039–1082.

20 Doron Avramov and Tarun Chordia. "Asset Pricing Models and Financial Market Anomalies". In: *Review of Financial Studies* 19.3 (2006), pp. 1001–1040.

21 Doron Avramov, Tarun Chordia, and Amit Goyal. "The Impact of Trades on Daily Volatility". In: *Review of Financial Studies* 19.4 (Jan. 2006), pp. 1241–1277.

22 X. Frank Zhang. "Information Uncertainty and Stock Returns". In: *Journal of Finance* 61.1 (2006), pp. 105–136.

23 Doron Avramov, Tarun Chordia, Gergana Jostova, and Alexander Philipov. "Momentum and Credit Rating". In: *Journal of Finance* 62.5 (Oct. 2007), pp. 2503–2520.

24 Joelle Miffre and Georgios Rallis. "Momentum Strategies in Commodity Futures Markets". In: *Journal of Banking and Finance* 31.3 (May 2007), pp. 1863–1886.

25 Eugene F. Fama and Kenneth R. French. "Dissecting Anomalies". In: *Journal of Finance* 63.4 (Aug. 2008), pp. 1653–1678.

26 Gary B. Gorton, Fumio Hayashi, and Geert K. Rouwenhorst. "The Fundamentals of Commodity Futures Returns". Yale ICF Working Paper No. 07-08. 2008.

27 Eli Beracha and Hilla Skiba. "Momentum in Residential Real Estate". In: *The Journal of Real Estate Finance and Economics* 43.3 (Oct. 2011), pp. 299–320.

28 Clifford S. Asness, Tobias J. Moskowitz, and Lasse H. Pedersen. "Value and Momentum Everywhere". In: *Journal of Finance* 68.3 (June 2013), pp. 929–985.

29 Doron Avramov, Tarun Chordia, Gergana Jostova, and Alexander Philipov. "Anomalies and Financial Distress". In: *Journal of Financial Economics* 108.2 (May 2013), pp. 139–159.

30 Gergana Jostova, Stanislava Nikolova, Alexander Philipov, and Christof Stahel. “Momentum in corporate bond returns”. In: *Review of Financial Studies* 26.7 (May 2013), pp. 1649–1693.